

Recap

- proof by contradiction
- If we want to show p is true
 - Assume $\neg p$ is true
 - Then, by deduction, we try to come up with a contradiction
 - $\neg p$ is false
 - p should be true.
- Direct proofs, proof by cases, proof by contradictions.

Proof by contrapositive

This is a technique that we can use to prove implications are true.

Let's consider an implication $p \Rightarrow q$

Then, its contrapositive statement is $\neg q \Rightarrow \neg p$

P	Q	$\neg P$	$\neg Q$	$P \Rightarrow Q$	$\neg Q \Rightarrow \neg P$
0	0	1	1	1	1
0	1	1	0	1	1
1	0	0	1	0	0
1	1	0	0	1	1

$$(P \Rightarrow Q) \equiv (\neg Q \Rightarrow \neg P)$$

Def

To give a proof by contrapositive of an implication $A \Rightarrow B$, we instead give a proof for $\neg B \Rightarrow \neg A$

claim: If it rains in Bozeman, then Bozeman is wet.

P : It rains in Bozeman

Q : Bozeman is wet.

contrapositive statement:

If Bozeman is not wet, then It does not rain in Bozeman.

Let $n \in \mathbb{Z}$,

claim: If n^2 is even, then n is even.

Since this is an implication, we can use proof by contrapositive to prove this.

P : n^2 is even.

Q : n is even.

contrapositive statement

If n is odd, then n^2 is odd.

$\equiv$
If n^2 is even, then n is even

Proof:

Assume n is odd

by assumption.

$$n = 2k + 1, k \in \mathbb{Z}$$

by def of odd

$$n^2 = (2k+1)^2 = 4k^2 + 4k + 1$$

by algebra.

$$n^2 = 2(2k^2 + 2k) + 1$$

by factoring

$$n^2 = 2c + 1, c = 2k^2 + 2k, c \in \mathbb{Z}$$

sum of prod. of
ints is an int

n^2 is odd

by def of odd.

We have shown that the contrapositive statement of the original implication is true.

∴ our initial claim is true.

Claim: All prime numbers are odd.

Disprove by counter example

2 is a prime number but not odd.

New claim: All prime numbers greater than 2 are odd.

Let $n \in \mathbb{Z}$, $n > 2$

If n is a prime number, then n is odd.

Premise (P) : n is a prime #

conclusion (Q) : n is odd.

contrapositive statement

If n is even, then n is ^{not} a prime #

Assume n is even

by assumption

$$n \in \mathbb{Z} \wedge n \geq 3$$

Given,

$$n = 2k, k \in \mathbb{Z}$$

by def of even

n is not a prime #

because $n > 2$,
and 2 is a factor of n .

The contrapositive statement is true,

$\therefore$ the original claim is true.

Claim: Given $x, y \in \mathbb{R}$

If $|x| + |y| \neq |x+y|$, then $xy < 0$

contrapositive statement

If $xy \geq 0$, then $|x| + |y| = |x+y|$

Assume $xy \geq 0$

by assumption.

Case 1: $x, y \geq 0 \Rightarrow xy \geq 0$

Case 2: $x, y < 0 \Rightarrow xy \geq 0$

Case 1: $x, y \geq 0$

$$xy \geq 0$$

because $xy \geq 0$

$$|x| + |y| = x + y$$

$x, y \geq 0$ and
by def of $|\cdot|$

$$x + y \geq 0$$

$$x, y \geq 0$$

$$|x + y| = x + y$$

by def of $|\cdot|$,
 $x, y \geq 0$
by substitution

$$|x| + |y| = |x + y|$$

Case 1 is proven.

case 2'. $x, y < 0$

$$xy > 0$$

because $x, y < 0$

$$|x| + |y| = -x + -y$$

$x, y < 0$ and
by def of $|\cdot|$

$$|x| + |y| = -(x+y)$$

by rearranging

$$x+y < 0$$

$$x, y < 0$$

$$-(x+y) > 0$$

by algebra

$$|x+y| = -(x+y)$$

by def of $|\cdot|$

$$|x| + |y| = |x+y|$$

by substitution

case 2 is proven.

We showed that contrapositive statement
is true for all cases.

$\therefore$ our original claim is true.

 